

LAB EXPERIMENT RECORD

Set 3: Employee Performance Evaluation

Department	B. Tech AI&DS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Dataset 1: Monthly Sales

Month	Sales (\$)
January	15,000
February	18,000
March	22,000
April	20,000
May	23,000

Dataset 2: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

4	Sales	4	90
5	HR	2	76

Program 1: Line Chart – Monthly Sales Trend

Aim

To create a line chart to visualize the monthly sales trend with appropriate labels and legend.

R Program

```

month      <-      c("January",      "February",      "March",      "April",      "May")
sales      <-      c(15000,      18000,      22000,      20000,      23000)

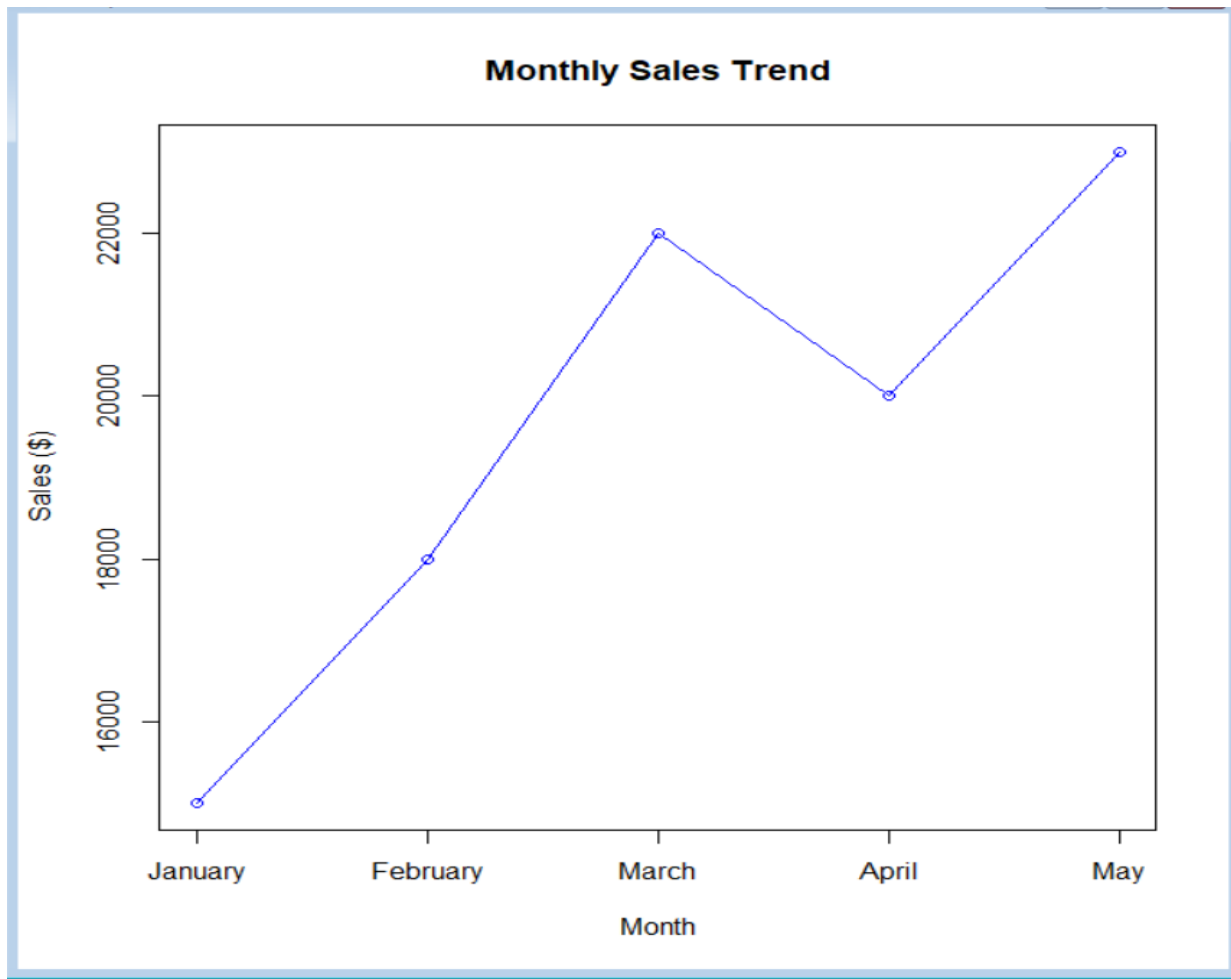
data              <-              data.frame(month,              sales)

plot(data$sales,
      type              =              "o",
      col              =              "blue",
      xaxt              =              "n",
      xlab              =              "Month",
      ylab              =              "Sales ($)",
      main              =              "Monthly Sales Trend")

axis(1,      at      =      1:5,      labels      =      data$month)

legend("topleft",
      legend              =              "Sales",
      col              =              "blue",
      lty              =              1,
      pch = 1)

```



Observation

Sales generally increased from January to May, with a small decrease in April. The highest sales were recorded in May at **\$23,000**.

Program 2: Bar Chart – Employee Distribution by Department

Aim

To create a bar chart showing the distribution of employees across different departments.

R Program

```
department      <-      c("Sales",      "HR",      "Marketing",      "Sales",      "HR")

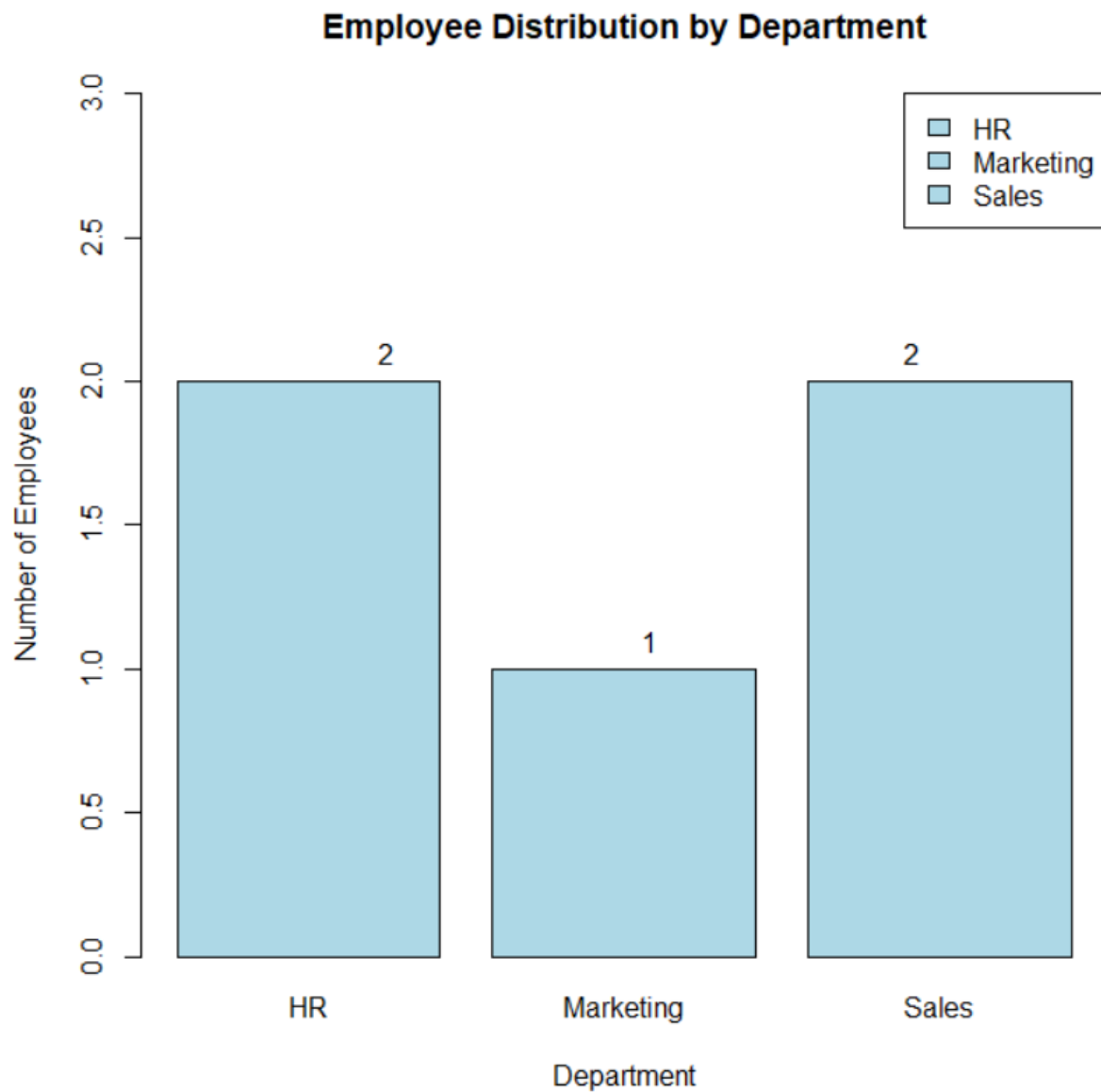
employee_data    <-      data.frame(department)

dept_count      <-      table(employee_data$department)

barplot(dept_count,
        main      =      "Employee      Distribution      by      Department",
        xlab      =      "Department",
        ylab      =      "Number      of      Employees",
        col      =      "lightblue",
        ylim      =      c(0,      max(dept_count)      +      1))

text(x            =      seq_along(dept_count),
     y            =      dept_count,
     labels       =      dept_count,
     pos         =      3)

legend("topright",
      legend      =      names(dept_count),
      fill = "lightblue")
```



Observation

Sales and HR each have **2 employees**, while Marketing has **1 employee**.

Program 3: Scatter Plot – Years of Service vs Performance

Aim

To create a scatter plot to analyze the relationship between employees' years of service and their performance scores.

R Program

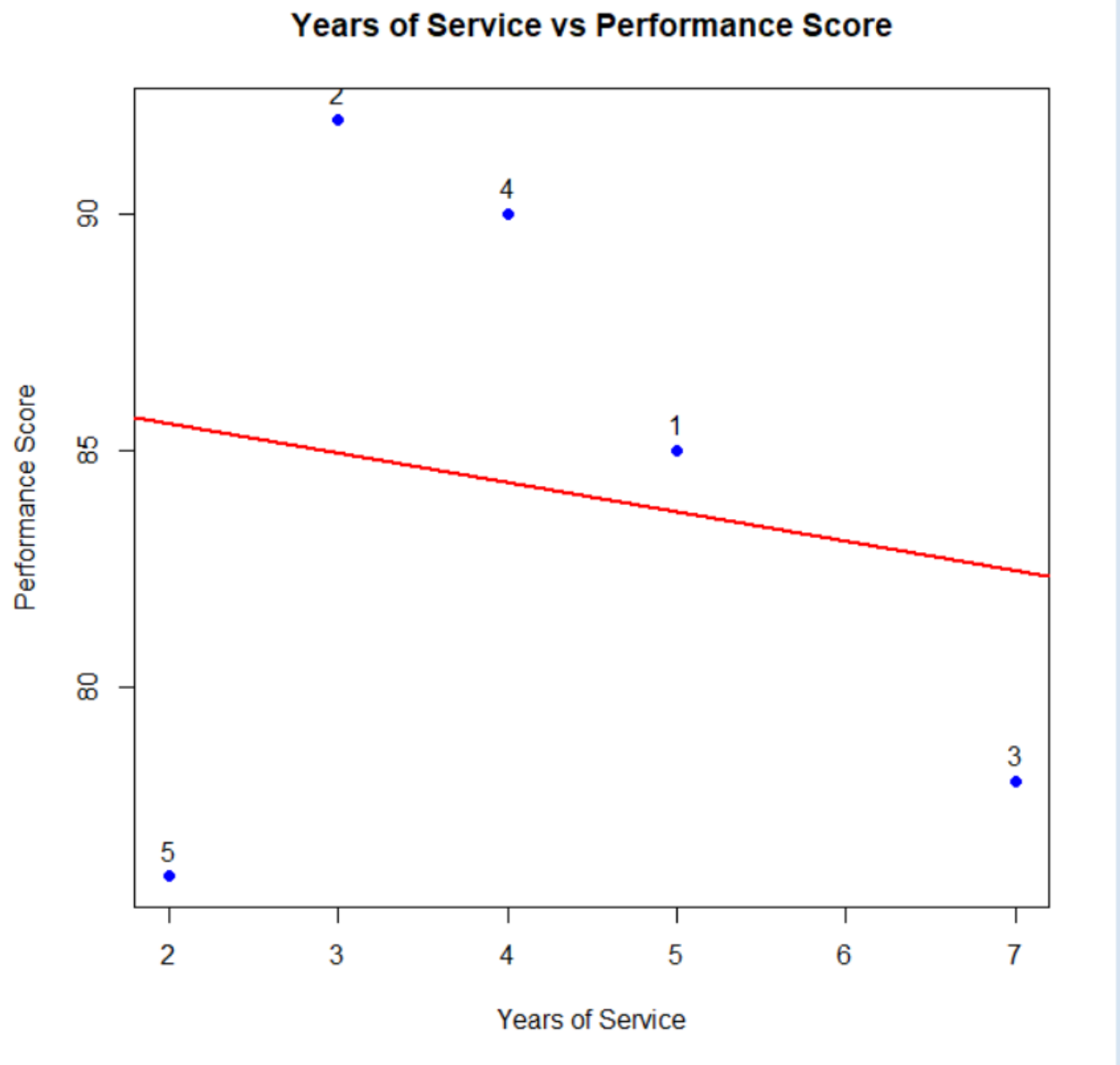
```
years      <-      c(5,      3,      7,      4,      2)
performance <-      c(85,      92,      78,      90,      76)

employee_data <-      data.frame(years,      performance)

plot(employee_data$years,
      employee_data$performance,
      main      =      "Years      of      Service      vs      Performance      Score",
      xlab      =      "Years      of      Service",
      ylab      =      "Performance      Score",
      pch      =      19,
      col      =      "blue")

text(employee_data$years,
      employee_data$performance,
      labels      =      1:5,
      pos      =      3)

abline(lm(performance ~ years, data = employee_data),
      col      =      "red",
      lwd = 2)
```



Observation

The scatter plot shows a **negative relationship** in this small dataset: employees with more years of service do not necessarily have higher performance scores. For example, the employee with 7 years of service has a score of 78, while an employee with 4 years has a score of 90.

Program 4: Tableau Dashboard

Aim

To develop an interactive Tableau dashboard containing the line chart and bar chart for employee performance analysis.

Procedure

Step 1: Open Tableau.

Step 2: Create the following datasets in Excel/CSV:

Sales Dataset

Month,Sales

January,15000

February,18000

March,22000

April,20000

May,23000

Employee Dataset

Employee ID,Department,Years of Service,Performance Score

1,Sales,5,85

2,HR,3,92

3,Marketing,7,78

4,Sales,4,90

5,HR,2,76

Step 3: Open Tableau → **Connect to Data** → select the Excel/CSV file.

Step 4: Create **Sheet 1 – Line Chart**

- Drag Month → Columns
- Drag Sales → Rows
- Select **Line** chart
- Add title: **Monthly Sales Trend**

Step 5: Create Sheet 2 – Bar Chart

- Drag Department → Columns
- Drag Employee ID → Rows
- Change aggregation to **Count**
- Select **Bar Chart**
- Turn on **Show Mark Labels**
- Add title: **Employee Distribution by Department**

Step 6: Create Dashboard

- Click **New Dashboard**
- Drag the Line Chart onto the dashboard.
- Drag the Bar Chart below or beside it.
- Add suitable title: **Employee Performance Evaluation Dashboard**

Step 7: Add filters/interactivity if required, such as Department.

Expected Dashboard

The dashboard allows users to visually explore:

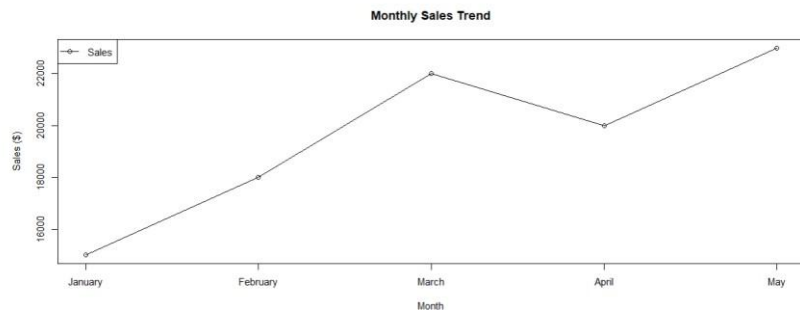
- Monthly sales trends
- Number of employees in each department
- Department-wise employee distribution
- Interactive filtering

Employee Performance Evaluation Dashboard

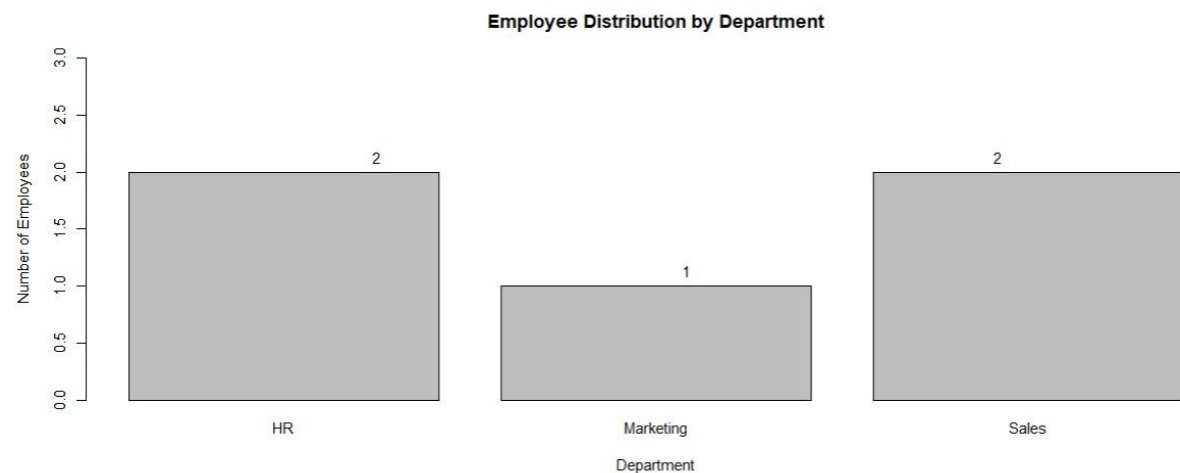
Select Department:

All

Monthly Sales Trend



Employee Distribution by Department



Overall Procedure

1. Create/import the required datasets.
2. Store the data in R data frames.
3. Create a line chart for monthly sales.
4. Create a bar chart for department distribution.
5. Create a scatter plot for years of service versus performance score.
6. Add labels, titles, legends, and trend line where required.
7. Import the datasets into Tableau.
8. Create the line and bar charts in Tableau.
9. Combine the charts into an interactive dashboard.
10. Analyze the resulting visualizations.

Result

The monthly sales trend, employee department distribution, and relationship between years of service and performance score were successfully visualized using **R**, and an interactive **Tableau dashboard** was designed using the line and bar charts.

Key R Functions Used

Function	Purpose
c()	Create vectors
data.frame()	Create data frames
plot()	Create line/scatter plots
axis()	Add/customize axes
legend()	Add legends
barplot()	Create bar charts
text()	Add labels to charts
table()	Count department frequencies
lm()	Create linear regression model
abline()	Add regression line